From Offline Optimality to Real-Time Simulation Flight: Delay-Aware LQR and Constrained DeePC for an EHEKATL Blended-Wing-Body Aircraft

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Abstract

Nominal optimal-control performance can deteriorate when a flight controller is transferred through delayed actuator routing, hidden augmentation and hard constraints. This study compares delay-aware linear-quadratic regulation (LQR) and constrained data-enabled predictive control (DeePC) for local roll-rate regulation of an EHEKATL blended-wing-body aircraft in X-Plane 12. Stabilised multisine tests identified the realised 35 Hz command-to-surface-to-rate path, while a recovery-capable backstepping controller provided permanent fallback. LQR reduced the matched integral error by 11.1%, below the predeclared 30% target. DeePC passed 243 of 243 model-uncertainty cases. Both controllers then executed safely at 35 Hz under common command and slew limits; DeePC's maximum solve time was 7.90 ms within a 28.57 ms period. Power analysis found that a superiority claim requires 48 pulses per controller under separate reloads or 24 validated paired pulses. The evidence supports real-time feasibility, but not DeePC superiority.

Keywords: blended-wing body; LQR; DeePC; data-driven control; real-time optimisation; flight simulation

1. Introduction

Linear-quadratic control is attractive for aircraft regulation because it provides a transparent state-feedback law and an explicit state-versus-control trade-off (Kalman, 1960). Its conclusions nevertheless depend on the fidelity of the model, state definition, actuator representation, delay model, and operating point. Data-enabled predictive control offers a complementary approach: rather than first reducing measured behaviour to a low-order parametric plant, DeePC uses input-output trajectories to predict and optimise future behaviour subject to constraints (Coulson et al., 2019, 2022). Whether this advantage survives real-time implementation is an experimental question.

The EHEKATL aircraft is a scale blended-wing-body model represented in X-Plane 12. Initial thesis-derived LTI studies produced stable LQR behaviour offline, but direct transfer to X-Plane was unsuccessful. Historical fixed-gain flights reached 53.0 deg absolute pitch and 111.9 deg absolute roll in one run, while a second run repeatedly activated divergence guards. These failures motivated a stage-gated redevelopment process based on measured actuator realization, independent training and validation data, immutable performance thresholds, and a recovery-capable backstepping baseline.

This paper makes five bounded contributions. First, it documents the implementation gap between nominal LQR analysis and the realised X-Plane control path. Second, it presents a stabilised identification procedure and a directly discrete 35-Hz roll cascade. Third, it implements constrained LQR and DeePC controllers behind a common safety, fallback, and actuator-restoration contract. Fourth, it measures online optimisation time against the control deadline. Fifth, it reports both successful real-time integration and failed performance and power gates without retrospectively weakening the criteria. The experiments are real-time closed-loop simulation flights; they are not physical-aircraft flight tests.

2. State of the art and research gap

Blended-wing-body (BWB) control research has progressed from local linear regulators towards integrated robust control, nonlinear inversion and allocation over many redundant surfaces. NASA and Boeing's X-48B programme is the main open flight-research reference. It combined envelope expansion, system-identification manoeuvres and a nonlinear dynamic-inversion architecture with pseudo-daisy-chain allocation across a highly redundant surface set (Glaab et al., 2009; Jordan et al., 2008). The programme also showed that instrumentation, surface realisation and an explicit envelope-expansion process are part of the controller experiment rather than peripheral implementation details.

Model-based work has extended LQR and LQG with mixed H2/H-infinity, structured H-infinity and linear-parameter-varying formulations. Denieul et al. (2017), for example, jointly considered BWB surface geometry, allocation and robust control-law design. These approaches retain interpretable robustness objectives, but their guarantees remain tied to the selected model, linearisation, scheduling variables and actuator representation. The broad 2025 BWB review by

Ali et al. and the tailless-aircraft review by Hu et al. (2024) identify stability, control coupling and redundant effectors as persistent configuration-level problems.

Control allocation is therefore central to tailless aircraft. Weighted pseudoinverse, direct allocation, redistributed pseudoinverse, quadratic programming and nonlinear or incremental allocation can distribute requested moments while respecting position and rate limits. Wind-tunnel BWB experiments show, however, that linear effectiveness matrices can lose material accuracy at large deflections and high incidence. Waters et al. (2013) reported typical moment losses of 10-30%, with some allocation cases delivering only about half the requested moment. That evidence motivates separate covariance-aware elevon-allocation work for EHEKATL, but the present paper keeps a single realised roll channel so that LQR and DeePC use exactly the same input and output.

Nonlinear and fault-tolerant BWB studies use backstepping, sliding-mode control, incremental methods and disturbance observers. Liu and Whidborne (2021) combined incremental backstepping, sliding-mode ideas and an RBF observer for fault tolerance. The present work uses backstepping in a different role: it is the recovery-capable baseline and permanent safety fallback, not a third optimiser in the performance contest.

DeePC follows the behavioural result that a persistently exciting trajectory can parametrise the finite trajectories of a controllable linear system (Willems et al., 2005). The original formulation and its distributionally robust extension established the optimisation framework (Coulson et al., 2019, 2022). Experimental aerial evidence includes quadrotors (Elokda et al., 2021), whereas open BWB studies remain dominated by model-based robust and nonlinear control. Hybrid DeePC now incorporates partial model knowledge to improve data efficiency and conditioning (Watson, 2025). Public evidence nevertheless remains sparse for DeePC acting through the delayed, augmented and surface-realisation path of a BWB simulator.

The defensible gap is therefore narrower than claiming a first use of DeePC in aircraft. Published DeePC flight-control evidence is concentrated in multirotors, while open BWB control work concentrates on model-based nonlinear, robust and allocation methods. A reproducible LQR-DeePC comparison through a measured BWB command-to-realised-actuator path, with common constraints, fallback logic, prospective power analysis and publication of failed gates, is underrepresented.

3. Aircraft, simulator, and operating condition

The test article was the EHEKATL BWB G33 scale aircraft in X-Plane 12.4.3-r2, interfaced through the NASA XPlaneConnect architecture (NASA Ames Research Center, n.d.). The frozen local operating condition, denoted R3, was approximately 1700 m altitude, 39 kt indicated airspeed, 240 Pa dynamic pressure, cockpit throttle 0.02353, 1.3 deg pitch, near-zero roll, and low body rates. Manual/Clear weather, no declared failures, enabled X-Plane artificial stability, disabled autopilot/autothrottle, live simulation speed, and replay off were required.

The common manipulated variable was trim-relative yoke-roll perturbation. Body roll rate P was the primary regulated output. Roll angle, total roll-control ratio, distributed aileron deflection, moments, simulator state, override ownership, and timing were retained as audit channels. Both optimized controllers ran at 35 Hz with command magnitude limited to 0.0002 normalised units and slew limited to 0.00005 per sample. The original backstepping law was updated at its validated 10-Hz rate and held between updates when used as the fallback.

4. Control-path diagnosis and identification

Early experiments showed that a transmitted XPlaneConnect CTRL packet did not necessarily equal the realised actuator input. When the simulator was paused or in replay, writable yoke values could change without advancing surfaces or aircraft dynamics. Even in live simulation, physical joystick input could overwrite nonpersistent commands between controller updates. Persistent per-axis ownership and direct yoke writes were therefore required.

Taking ownership exposed an additional trim problem. The original controller implicitly relied on a physical pitch input near 0.149 normalised units. A first owned-axis handoff removed this contribution and produced a pitch-rate transient of approximately -13.85 deg/s. Subsequent tests captured the realised trim before ownership and applied controller outputs as trim-relative perturbations. Artificial stability remained active and altered the mapping between yoke command and total control; consequently, measured total control, not sent command alone, was used for plant identification.

Stabilized multisine experiments were superimposed on backstepping rather than replacing it. Two training and two held-out validation runs were collected for each of roll and pitch. All eight datasets passed completeness, realization,

spectral-richness, safety, and restoration gates. A 10-Hz local angle/rate model identified a three-sample input delay but retained colored residuals. Higher-rate testing subsequently isolated the measured cascade

$$u_{y,k} \rightarrow u_{t,k} \rightarrow \delta_{a,k} \rightarrow P_k$$

where commanded yoke perturbation maps dynamically to total roll control, total control maps almost statically to distributed aileron motion, and surface motion maps dynamically to roll rate. Temporal held-out validation supported a directly discrete 35-Hz cascade, while continuous-time pole interpretations were rejected because fitted time constants were at or below the sampling interval.

5. Delay-aware LQR

The frozen roll-rate LQR state was

$$x_k = [\Delta P_k,\ \Delta u_{t,k},\ \Delta u_{y,k-1}]^T$$

with the identified dynamics

$$x_{k+1} = Ax_k + Bu_k$$

The discrete matrices were

$$A = \begin{bmatrix} 0.40856 & 3.55738 & 37.94551 \\ 0 & 0.05609 & 0.59835 \\ 0 & 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 23.90120 \\ 0.37689 \\ 1 \end{bmatrix}$$

The input minimized the infinite-horizon quadratic cost

$$J = \sum_{k=0}^{\infty} (x_k^T Q x_k + u_k^T R u_k)$$

subject at runtime to the common command and slew limits. The resulting gain was

$$K = [0.0002944,\ 0.0027424,\ 0.0292528]$$

The nominal closed-loop spectral radius was 0.3684. Across 243 coefficient combinations, the worst spectral radius was 0.4936 with no unstable cases. These results establish robustness of the frozen measured-cascade model, not global stability of the X-Plane aircraft.

6. Regularised constrained DeePC

The selected DeePC used one input (trim-relative yoke-roll perturbation) and one output (roll-rate perturbation). Attempts to predict P, roll angle, and total roll control jointly failed the reconstruction gate; the final claim is therefore explicitly P-only. With Hankel matrices constructed from the frozen design trajectory following the behavioural trajectory parameterisation (Coulson et al., 2019; Willems et al., 2005), each control update solved a regularised finite-horizon problem of the form

$$\min_g \ \|y_f\|_{Q_y}^2 + \|u_f\|_{R_u}^2 + \|\Delta u_f\|_{R_{\Delta u}}^2 + \lambda_g \|g\|_2^2$$

subject to past input/output consistency, future behavioural constraints, and hard command/slew bounds. The frozen configuration used five past samples, five prediction samples, \(\lambda_g=10\), \(Q_y=1\), \(R_u=0.01\), and \(R_{\Delta u}=0.02\). Constant optimiser matrices were precomputed, reducing online runtime without changing commands; packaged-versus-design command error was exactly zero.

The controller contained a permanent fallback latch. Nonfinite inputs, optimiser nonconvergence, deadline violation, simulator-state violation, or safety-envelope violation permanently transferred selection to bounded backstepping for the remainder of the run. Automatic DeePC re-engagement was not permitted.

7. Experimental protocol and metrics

Every engaged run first evaluated altitude, airspeed, pitch, roll, body rates, artificial-stability state, override ownership, pause, simulation speed, and replay mode. A failed start gate sent no control. After a passing gate, roll and pitch axes were owned explicitly. Pitch remained stabilised by the original backstepping law, while the tested controller supplied the roll perturbation. Captured yoke values were restored in a finally path and verified after the run.

The primary performance measure was time-integrated absolute roll-rate deviation. Secondary measures included peak and final-window roll-rate deviation, control energy, command and slew utilization, computation time, deadline misses, fallback events, and restoration error. Performance thresholds and dataset roles were frozen before confirmation data were opened. A safe run was treated as integration evidence, not automatically as a performance pass.

8. Results

8.1 LQR matched-performance result

The measured-cascade LQR passed its integration and safety prerequisites. In the frozen matched backstepping/LQR flight pair, the final integral ratio was 0.888807, or an 11.1% reduction relative to backstepping. The predeclared gate required a ratio no greater than 0.70. The LQR performance claim therefore failed. The ratio worsened as the quiet tail was accumulated, indicating that reload-specific background motion dominated the small disturbance response.

8.2 DeePC offline robustness and integration

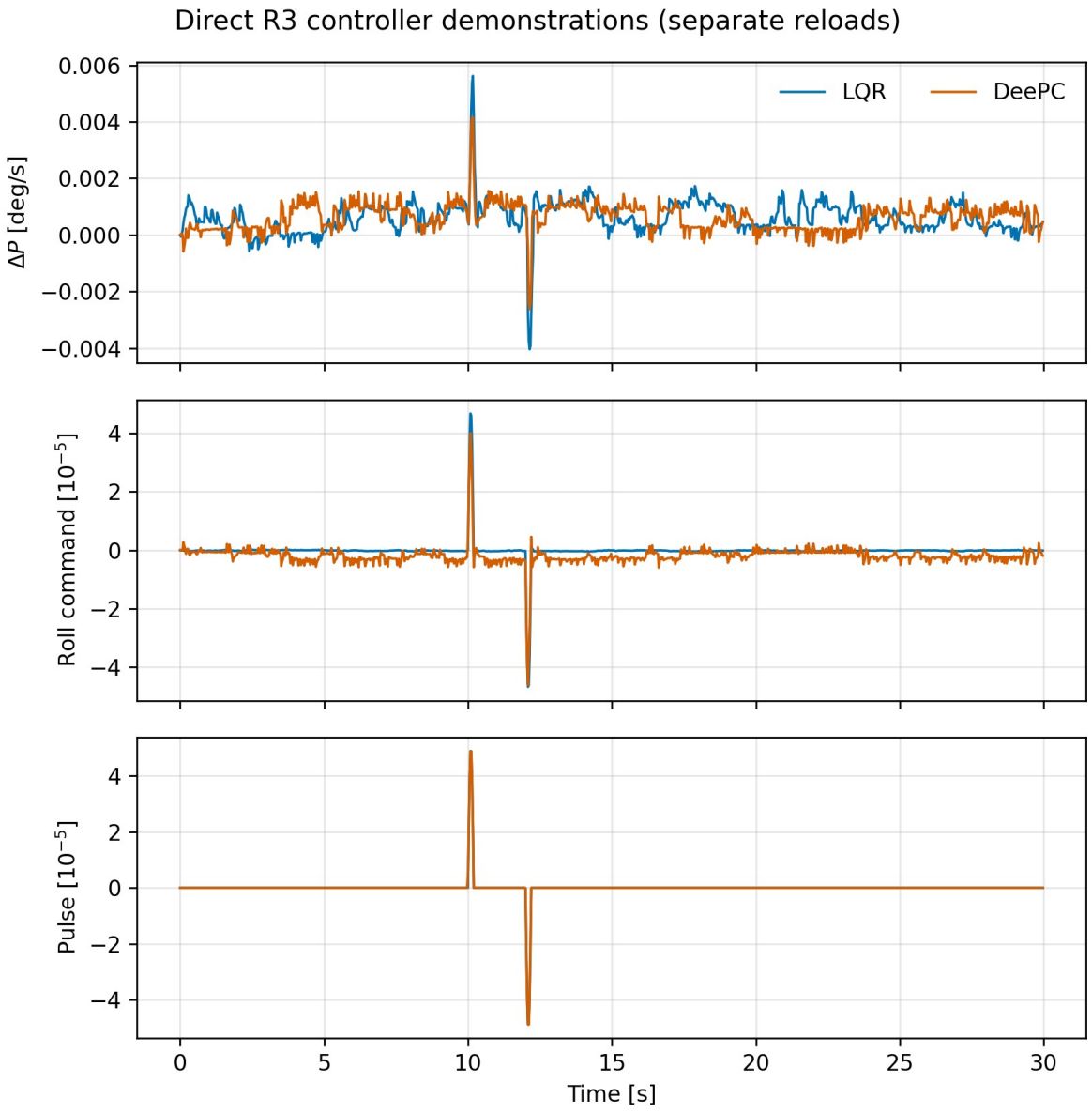
The final DeePC candidate converged in all 243 frozen-model uncertainty cases. The worst and median DeePC/LQR integral ratios were 0.895344 and 0.788056, and the worst peak ratio was 0.928154. Maximum command and base slew were 4.39e-5 and 9.97e-6, respectively, within the common constraints. A guarded design pilot passed all integration gates and was excluded from confirmation.

8.3 Direct demonstrations

Table 1 reports the two user-requested direct demonstrations. Each began after a fresh R3 reload; consequently, the values are descriptive and are not a matched comparison.

Metric	DeePC	LQR
Samples	1050	1050
Effective frequency [Hz]	35.0014	35.0040
Maximum gap [ms]	50.09	42.23
Integral absolute delta-P [deg]	0.01972	0.02025
RMS delta-P [deg/s]	0.000796	0.000863
Peak absolute delta-P [deg/s]	0.004163	0.005622
Final-5-s maximum absolute delta-P [deg/s]	0.001311	0.001496
Peak command	4.609e-5	4.670e-5
Maximum slew	2.126e-5	2.171e-5
Fallback/safety event	none	none
Verified restoration	pass	pass

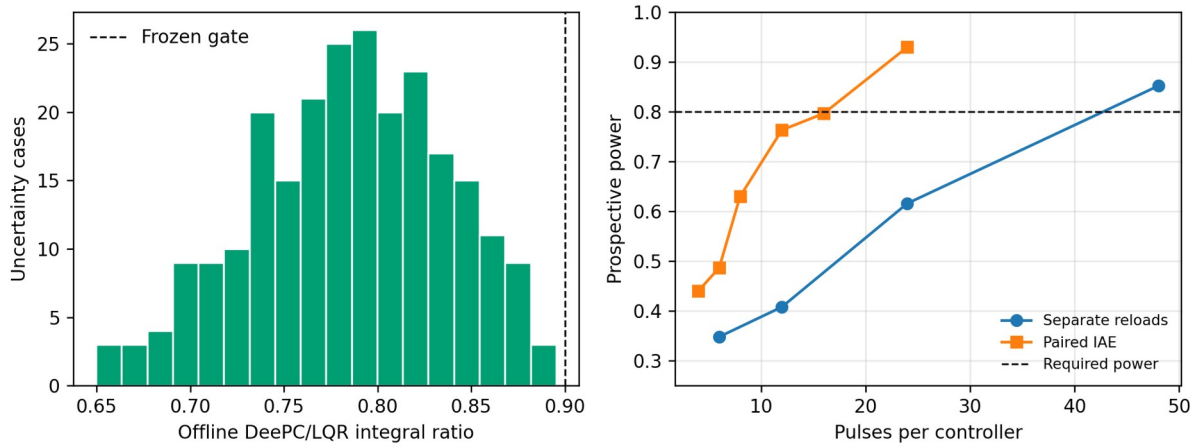
DeePC median, 95th-percentile, and maximum optimisation times were 4.02, 5.36, and 7.90 ms, respectively, below the 28.57-ms sample period. Both controller runtimes reproduced their frozen offline behaviour exactly. The slightly smaller DeePC response metrics cannot be interpreted causally because the initial reloads and background trajectories were not matched.



Direct controller demonstrations

8.4 Prospective power

Prospective moving-block bootstrap analyses quantified the cost of an independent superiority claim. A separate-reload design first exceeded 0.80 power at 48 pulses per controller, requiring at least 149 s per controller. A paired mean per-pulse IAE design achieved power 0.7967 at 16 pairs, which failed the unchanged 0.80 gate; it first qualified at 24 pairs and at least 152 s. That design also required controller switching and history handoffs not yet validated in flight. The benchmark was therefore rejected before collection.



Offline robustness and prospective power

9. Discussion

The experiments show why offline optimality and online performance should be treated separately. The LQR was stable and robust over its identified model, yet its matched improvement was small in X-Plane. The dominant complications were not numerical Riccati-solver limitations: they were control ownership, hidden trim, artificial-stability dynamics, a dynamic commanded-to-realised input path, local model validity, and reload-specific background motion.

DeePC reduced dependence on an explicit low-order state realization and passed the frozen offline robustness gate. It also met the real-time computational budget comfortably. Nevertheless, using measured data does not eliminate distribution shift. The selected controller predicts only P, its design data come from a local R3 trajectory, and the effect expected over LQR was too small relative to observed variability for an economical confirmation campaign.

The negative gates are an engineering result. Relaxing the 30% LQR target after observing an 11.1% improvement, rounding 0.7967 power to 0.80, or presenting separate-reload demonstrations as a matched comparison would each overstate the evidence. Preserving these failures yields a reproducible account of what the controllers do reliably: bounded local roll-rate regulation with verified real-time execution, fallback readiness, and restoration.

10. Limitations

The conclusions apply only near R3 and do not establish transfer to other airspeeds, dynamic pressures, mass properties, weather conditions, or failure states. Artificial stability was enabled and forms part of the tested plant. The work validates roll-rate behaviour, not a complete attitude loop, combined longitudinal/lateral operation, or large-disturbance recovery. DeePC is P-only. The direct demonstrations are single, unmatched runs. Finally, X-Plane is a high-fidelity simulation environment, not physical flight hardware; network, sensor, actuator, and aerodynamic effects on a real aircraft may differ.

11. Conclusions

A delay-aware measured-cascade LQR and a regularised constrained DeePC were implemented successfully for the EHEKATL BWB in real-time X-Plane flight. Both direct demonstrations met timing, command, slew, safety, exact-replay, fallback, and actuator-restoration requirements. DeePC optimisation remained well within the 35-Hz deadline. However, LQR failed its predeclared 30% matched-improvement gate, and a powered DeePC-versus-LQR superiority experiment was impractical under the available local protocol. The appropriate conclusion is therefore not that one controller universally outperforms the other, but that safe implementation is achievable while predicted performance differences can be overwhelmed by the realised control path and operating variability. Future work should prioritize multi-regime data, state/augmentation measurements, gain-scheduled or multi-dataset formulations, and a prospectively powered common benchmark before extending to attitude control.

Data and reproducibility statement

Every paper-facing dataset is listed with its role and SHA256 hash in paper/DATASET_REGISTRY.csv. Raw logs are retained under logs/, frozen numerical artifacts under configs/, and controller/analysis code under lqr/, deepc/, and

scripts/. Paper tables and figures are regenerated by paper/generate_paper_results.py. Design, demonstration, and held-out evidence are kept distinct. No physical-aircraft data are claimed.

Generative AI use disclosure

Generative AI tools were used for language editing, literature organisation and document preparation. The authors verified the calculations, source records, software, experimental evidence and claim boundaries and retain full responsibility for the manuscript.

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